

Questions with Answer Keys

MathonGo

Q1: 24 Feb (Shift 1) - Single Correct

Given below are two statements :

Statement I : Colourless cupric metaborate is reduced to cuprous metaborate in a luminous flame.

Statement II : Cuprous metaborate is obtained by heating boric anhydride and copper sulphate in a non-

luminous flame. In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Statement I is false but statement II is true.

(2) Statement I is true but Statement II is false.

(3) Both Statement I and Statement II are true.

(4) Both Statement I and Statement II are false.

Q2: 24 Feb (Shift 2) - Numerical

Among the following allotropic forms of sulphur, the number of allotropic forms, which will show paramagnetism is

(1) α -sulphur

(2) β -sulphur

(3) S_2 -form

Q3: 25 Feb (Shift 1) - Single Correct

The correct statement about B_2H_6 is:

(1) All B-H-B angles are of 120° .

(2) Its fragment, BH_3 , behaves as a Lewis base.

(3) Terminal B-H bonds have less p-character when compared to bridging bonds.

(4) The two B — H — B bonds are not of same length.

Q4: 25 Feb (Shift 1) - Numerical

Questions with Answer Keys

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Among the following, the number of halide(s) which is/are inert to hydrolysis is

- (A) BF_3
- (B) SiCl_4
- (C) PCl_5
- (D) SF_6

Q5: 25 Feb (Shift 2) - Single Correct

Given below are two statements :

Statement I : α and β forms of sulphur can change reversibly between themselves with slow heating or slow cooling.

Statement II : At room temperature the stable crystalline form of sulphur is monoclinic sulphur.

In the light of the above statements, choose the correct answer from the options given below.

- (1) Both statement I and statement II are false
- (2) Statement I is true but statement II is false
- (3) Both statement I and statement II are true
- (4) Statement I is false but statement II is true

Q6: 25 Feb (Shift 2) - Single Correct

Which of the following compound is added to the sodium extract before addition of silver nitrate for testing of halogens ?

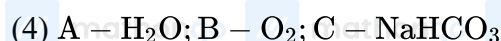
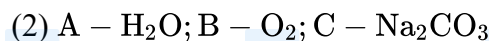
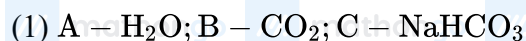
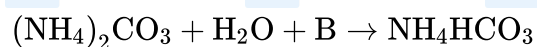
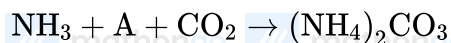
- (1) Nitric acid
- (2) Sodium hydroxide
- (3) Hydrochloric acid
- (4) Ammonia

Q7: 26 Feb (Shift 1) - Single Correct

Questions with Answer Keys

MathonGo

Find A, B and C in the following reaction :

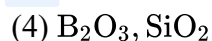
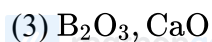
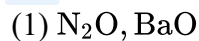
**Q8: 26 Feb (Shift 1) - Single Correct**

Compound A used as a strong oxidizing agent is amphoteric in nature. It is the part of lead storage batteries.

Compound A is :

**Q9: 26 Feb (Shift 2) - Single Correct**

Which pair of oxides is acidic in nature?

**Q10: 26 Feb (Shift 2) - Single Correct**

Questions with Answer Keys

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Match List-I with List-II.

List-I

(a) Sodium Carbonate

(b) Titanium

(c) Chlorine

(d) Sodium hydroxide

List-II

(i) Deacon

(ii) Caster-Kellner

(iii) Van-Arkel

(iv) Solvay

Choose the correct answer from the option given below:

(1) (a) – (iii), (b) – (ii), (c) – (i), (d) – (iv)

(2) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)

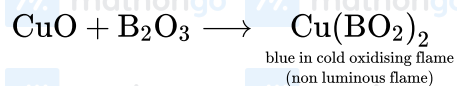
(3) (a) – (iv), (b) – (i), (c) – (ii), (d) – (iii)

(4) (a) – (i), (b) – (iii), (c) – (iv), (d) – (ii)

Answer Key**Q1 (4)****Q2 (1)****Q3 (3)****Q4 (1)****Q5 (2)****Q6 (1)****Q7 (1)****Q8 (2)****Q9 (4)****Q10 (2)**

Q1 (4)

Both are False

(1) Copper sulphate form copper meta boric with beric an hydride $\text{CuSO}_4 \longrightarrow \text{CuO} + \text{SO}_3$ 

(2) Blue coloured metal borate is reduced to copper in a luminous flame.

Q2 (1)

 S_2 is like O_2 ; e paramagnetic as per molecular orbital theory.

Q3 (3)

Terminal bond angle is greater than that of bridge bond angle Bond angle

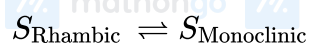
Bond order $\propto$ S-character

$$\propto \frac{1}{p\text{-character}}$$

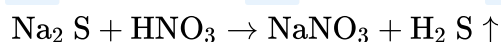
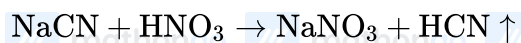
Q4 (1)

Due to crowding SF_6 is not hydrolysed.

Q5 (2)

 α -sulphur 95.6°C β -sulphur

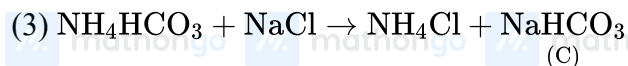
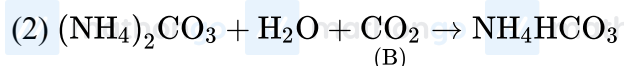
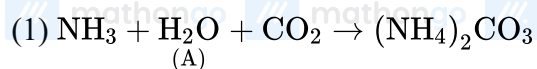
Q6 (1)

Nilnic acid decomposed NaCN & Na_2S , else they precipitate in test & misquite the resolve

Q7 (1)

Hints and Solutions

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Q8 (2)

lead storage batteries PbO_2 is used. In this O.S. of Pb is +4 so it is always reduced and behaves as oxidizing agent

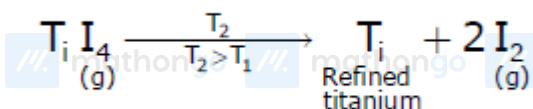
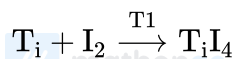
Q9 (4)

B_2O_3 and SiO_2 both are oxides of non-metal and hence are acidic in nature.

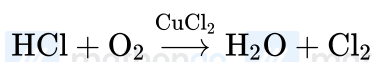
Q10 (2)

Sodium carbonate Na_2CO_3 & NaHCO_3

Titanium : Van arkel method



Chlorine : Decon's process



Sodium hydroxide :- Caster-Kellner cell